NRK Air Services

AI-Enabled Geo-Vision Aerial Logistics for Smart last-mile Delivery and Emergency HealthCare Transport across India.



Abstract: Why Air Service Matters

Challenge

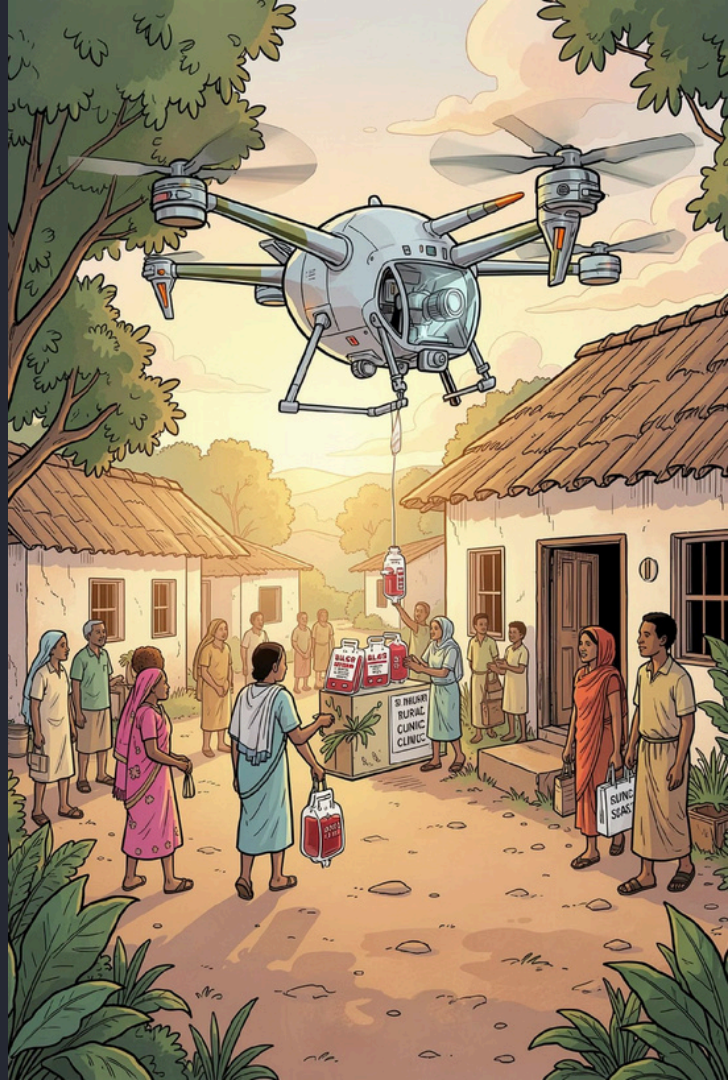
Road dependence → delays, high fuel use, limited access to remote and disaster zones.

Solution

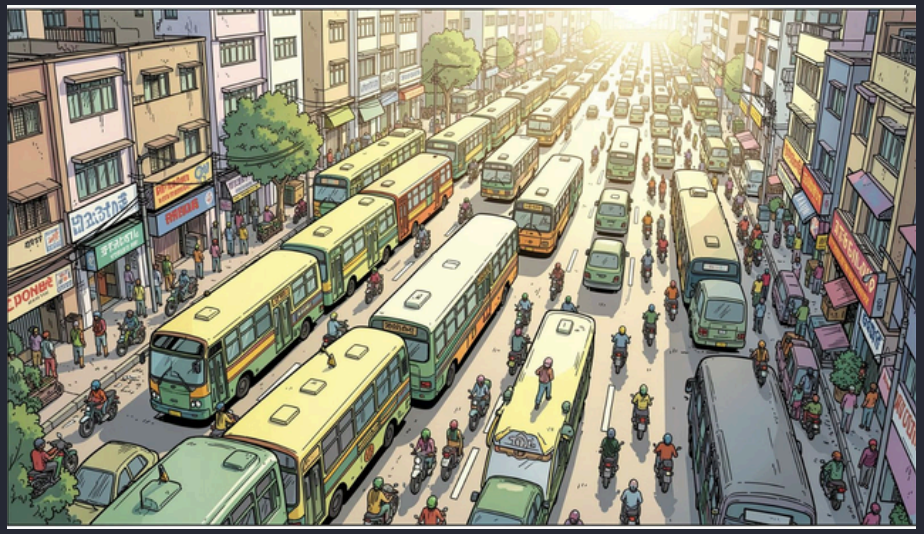
Integrate AI, computer vision, geospatial intelligence, autonomous drones, and cloud fleet management.

Impact

Faster deliveries, lower costs & emissions, improved emergency healthcare access.



Introduction: Context in India



Urbanization and traffic congestion strain traditional logistics.

Fuel dependency raises costs and emissions.

GPS-only drone navigation struggles in dense urban areas, villages, apartment complexes, and disaster zones where visual verification is crucial.

Problem Statements

PS-1
Traffic-induced delays reduce last-mile and emergency delivery efficiency.

PS-2
GPS-only navigation lacks intelligent visual verification for precise deliveries.

PS-3
High fuel & labor dependency increases costs and carbon emissions.

PS-4
Remote, rural, and disaster-affected areas face delayed access to critical supplies.



Literature Review (2023–2026)



AI-Driven Routing (2023)

Dynamic route planning improves delivery efficiency.



Healthcare Drones (2023)

Autonomous networks reduced emergency medical delivery times.



Vision-Based Localization (2024)

Visual land mark recognition outperforms GPS-only localizatio

n.



Smart City Mobility (2024)

Proposed drone corridors and urban aerial frameworks.



Edge AI (2025)

Onboard processing enables real-time decisions and obstacle avoidance.



Collaborative Swarms (2025)

Multi-agent coordination improves task allocation for many deliveries.



Geo-Vision Models (2026)

Satellite+visual matching improves localization accuracy.

Review Analysis

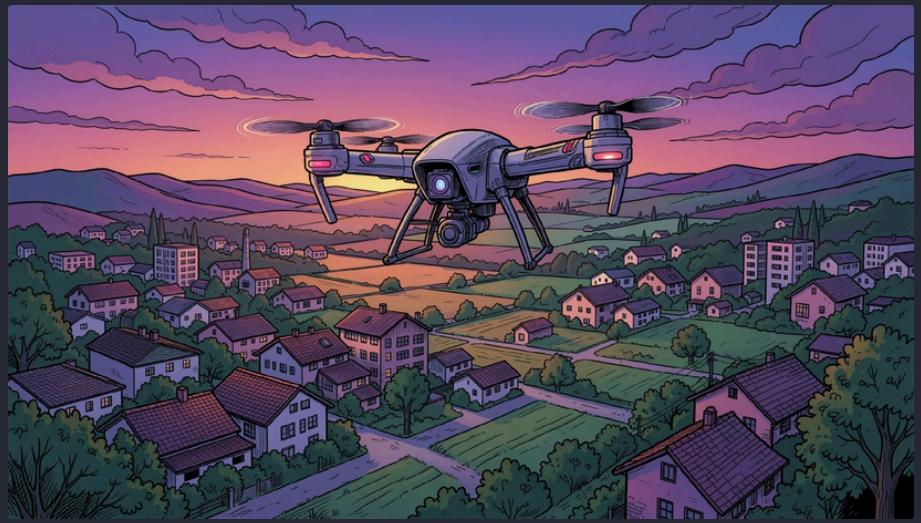
- **Cross-View Geo-Localization (University-1652, VIGOR):** Established UAV to ground matching datasets and metric learning approaches, but limited to Western urban environments with structured addressing systems, leaving Indian informal settlements unaddressed.

- **CLIP & Vision-Language Models for Localization:** Recent works apply contrastive vision-language embeddings for place recognition, but focus on single-modality matching; no work fuses text descriptions + ground image + GPS into a unified delivery verification pipeline.

- **UAV Last-Mile Delivery Systems:** Studies on autonomous drone delivery (Zipline, Wing) demonstrate operational feasibility but rely purely on GPS coordinates, lacking visual confirmation mechanisms for GPS-denied or address-ambiguous zones.

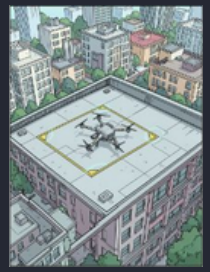
- **Indian Urban Scene Understanding:** Limited computer vision research exists on Indian streetscapes (IDD dataset for driving); no dedicated dataset or model targets aerial-to-ground matching for Indian rooftop/landmark features specifically for logistics applications.

Research Gap



- **No India-Specific Cross-View Dataset:** Existing datasets cover Western cities; Indian rooftops, compound walls, and landmark-based addresses are absent.
- **No Multimodal Address Verification:** No system combines ground photo + text description + GPS for delivery confirmation.
- **GPS-Denied Final-Meter Problem Unsolved:** Current drone delivery assumes GPS sufficiency; rural and dense urban zones remain unaddressed.
- **No Edge-Offload Verification Architecture:** All existing models target high-compute drones; ESP32-class offloaded pipelines are unexplored.

Novelty: What NRK Adds



1

Trimodal fusion pipeline

Combining ground photo + text landmark description + GPS for delivery location verification — a first in last-mile logistics.

2

GPS-denied delivery confirmation

Using visual-language matching as fallback for rural, dense urban, and disaster zones.





3

India-specific cross-view dataset

Capturing rooftops, compound walls, water tanks, and landmark-based addresses for aerial-to-ground matching.

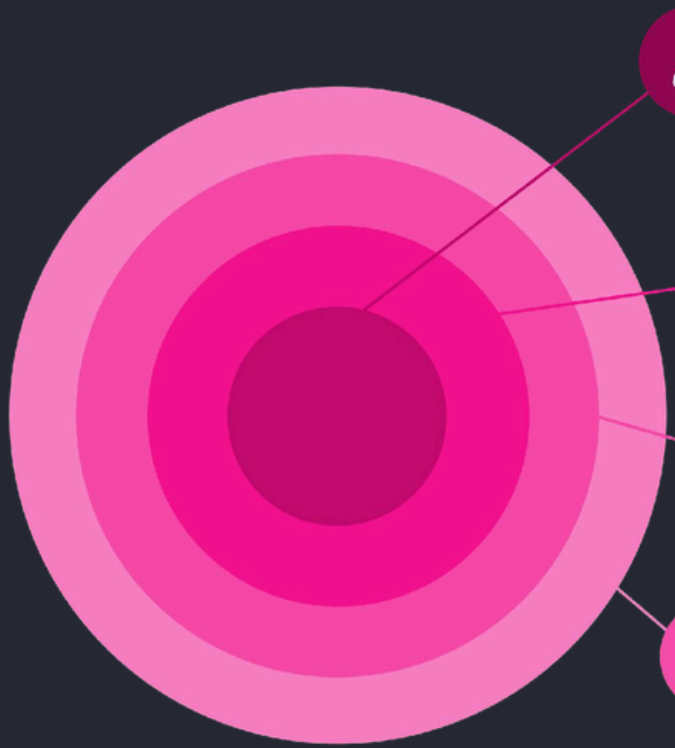


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Lightweight edge-offload

Architecture demonstrating cross-view matching on ESP32 drone hardware via server offloading.

Proposed Architecture



Core User Layer

Mobile app,order placement System, image upload module, Real-Time Delivery Tracking

Intelligence Layer

AI vision engine, geo-vision verification system, landmarks recognition module, route optimization module

Air Operations Layer

NRK Autonomous Drone Fleet, Flight Control System, Obstacle Detection and Avoidance, Smart Landing Zone Detection, Delivery Validation Module

Cloud Management Layer

NRK Central Command Centre, Vendor Management Platform, Fleet Monitoring Dashboard, Analytics and Reporting Engine



Expected Outcomes

Delivery Performance
Reduction in delivery time by up to 70%.

Operational Gains
Lower fuel dependency, reduced operational costs, scalable fleet architecture.

Healthcare Impact
Faster transport of emergency medicines, blood, and vaccines.

Sustainability
Reduced carbon emissions and promotion of greener last-mile logistics.

Accuracy & Access
Improved delivery accuracy via Geo-Vision; greater rural accessibility.

Future Vision
Next-gen AI aerial logistics ecosystem tailored to India.